

Python list

A single or group of elements which are enclosed in between square brackets and separated with comma are called list in Python.

→ For example $x = [1, 2, 3, 4]$

→ How to create single element list

$l = [10]$

→ How to create multi elements list with comma separator

$l = [10, 20, 30]$

→ Homogenous list object -- $[10, 20, 30, 40, 50]$

→ Heterogenous list object -- $[10, 20.50, \text{True}, 40, [1, 2, 3]]$

→ List datatype represents using 'list' type

$lst = [1, 2, 3, 4]$

→ we can create empty list object by using empty square brackets or list() function

for example $l1 = []$ $l2 = \text{list}()$

→ List is mutable type object

* List follows insertion order

* every character in the list object is indicated with unique index

→ Python support both forward and backward index

* Forward index start from 0 and it is representing first element as given object

→ Backward index start from -1 and it is representing last element as given object

* Python list supports "concatenation behaviour"

--> $[10, 20] + [30, 40]$

$[10, 20, 30, 40]$

--> Python list supports Repetition behaviour

$[10, 20] * 3$

$[10, 20, 10, 20, 10, 20, 10, 20]$

--> Python list supports List unpacking (Extracting list elements)

$a, b, c = [10, 20, 30]$

a

10

b

20

c

30

Python list datatype methods

object.count (element)

object.index (element, start-index=0, end-index=len(object)-1)

① How to add new elements into given list objects?

object.append (object)

object.extend (iterable object)

object.insert (index, object)

Exo- lists.append(10)

list2

= 10

Extend
lst2 = Extend ('python') — iterable object
one by one adding

[10, 'p', 'y', 't', 'h', 'o', 'n']

* Famous Interview question

what is the difference between append and
Extend

lst2 = append (10)

lst2

{10}

lst2 = extend ('python')

lst2

[10, 'p', 'y', 't', 'h', 'o', 'n']

insert

lst3 = [8, 4, 1, 5, 2, 9]

lst3.insert (1, 3)

output [8, 3, 4, 1, 5, 2, 9]

10) How to Remove existing elements from Given objects ?

object.remove(value) — should be compulsory
(It takes one argument only)

object.pop(index = -1) — If not given value error showing

object.clear()

del object

del object {indexing / slicing}

Remove

+ we need to pass one argument compulsory

If we not pass then Name error throwing

Pop

If we don't pass last element deleting
no value error ✓

Exo- Lst 3 [3, 4, 1, 5, 2, 9, 10, 4]

lst3.pop()

output Lst 3 [3, 4, 1, 5, 2, 9, 10]

clear

clear method is used to clear all elements which are
in the list. But memory there

Exo-

lst2

['h', 'o', 'n', 20, 30]

del lst2

entire deleting including object also

* object.sort

* By using sort method we can arrange the elements in ascending or descending order

Exo-

lst1 = [20, 40, 10, 30]

lst2 = sorted(lst1)

lst1

Display original object [20, 40, 10, 30]

lst2 = [10, 20, 30, 40]

Important

Point

* * *
IP your Requirements in the same
objects you want to change sort method

IP you want to change the main
function you have go for sorted
function

Reverse()

* IF we need to Reverse the elements in the same memory we use reverse()

Ex:-

lst =

[40, 30, 20, 10]

lst.reverse()

output [10, 20, 30, 40]

Reversed Function()

IF I donot want to do changes in main object and Return new object we have to for this

Ex:-

lst =

[10, 20, 30, 40]

reversed(lst)

Output

↳ new object

* IF you need lst

lst (reversed(lst))

output

[10, 20, 30, 40]

Copy

~~also~~

* Copying the value to another object

Exo

lst1

[15, 20, 30, 40]

lst3 = lst1.copy()

output

lst1 = [15, 20, 30, 40] lst3 [15, 20, 30, 40]

* Inside

list

Immutable type and mutable allowed